MST BLOCKCHAIN BUILDATHON #2 · ROUND 2 MVP

PatchPool

Fund the fix. Lock the commit. Release the reward.

An on-chain escrow and review workflow that links a software-fix bounty to a specific code submission, a human reviewer, and a recorded settlement.

MST MAINNET · 4646 | VERIFIED DEPLOYMENT | LOCAL MVP TESTS PASS

Team lead: Shravani Mankar · Team member listed in project brief: Vedika Ukirde

THE PROBLEM

Funding, review, and payment lose their link

Software-fix bounties often coordinate the task, source revision, reviewer decision, and reward across different tools. Participants may struggle to see what was reviewed, who was authorized to decide, and where escrow went.

Before PatchPool

Task board → code host → informal review → separate payment

Records are fragmented, and payment conditions may be enforced manually.

PatchPool's scope

Link the funded task to a committed revision, evidence digest, reviewer decision, and settlement recipient.

PatchPool complements Git hosting and human review; it does not replace them.

THE WORKFLOW

One contract-governed bounty lifecycle

01 · Sponsor

Create a task, assign reviewer, fund escrow; other sponsors may add funds.

02 · Contributor

Submit Git commit ID, evidence digest, and payout address.

03 · Reviewer

One assigned human reviews the specific submission and approves or rejects.

04 · Settlement

Approved reward goes to the stored payee; release can be retried.

05 · Expiry

Each eligible contributor claims their own refund after deadline + 7 days.

> The smart contract enforces escrow, address roles, and settlement. A human reviewer judges code quality.

ARCHITECTURE

Lean stack, clear trust boundaries

Application

Next.js · React · TypeScript

Wagmi · Viem · TanStack Query

BridgeKey-identified injected wallet

Mainnet production chain list is 4646 only. Every write needs normal wallet confirmation.

Contract

Solidity 0.8.20 · OpenZeppelin ReentrancyGuard · Hardhat

Per-wallet contribution accounting, submission records, reviewer decision, payee-bound payout, individual refunds.

Read path

Browser → same-origin bounded JSON-RPC proxy → MST Mainnet RPC

Proxy forwards allowlisted reads; transaction broadcast and signing methods are rejected.

Evidence boundary

Task JSON is public and hash-bound. Git object ID and evidence digest identify submitted values.

Evidence files stay off-chain. A matching digest proves byte equality only—not truth or correctness.

CONTRACT GUARANTEES AND LIMITS

Rules are enforced at the address level

Enforced

- ✓ Creator and contributors cannot submit or receive payout.
- ✓ Reviewer cannot fund, submit, or receive payout.
- ✓ Approval is bound to one submission ID.
- ✓ Reentrancy-protected payout and pull refunds.
- ✓ No refund after approval; eligible claims begin after seven days.

Not guaranteed

- Code quality, tests, or evidence truth.
- Distinct human identities behind addresses.
- Reviewer impartiality; one reviewer only.
- Quorum, arbitration, or dispute resolution.
- Independent professional audit.

PRODUCTION DEPLOYMENT

Verified on MST Mainnet

NETWORK

MST Mainnet · 4646

Deployment block

21,783,933

CONTRACT ADDRESS

`0x707004013b56e36eE458966d928539ede6166338`

Runtime Keccak-256

`0x3fd747f5a8d30110c3c75bf33d039dca659b07cd6b064b52cb861f25b5419aaf`

Successful deployment transaction: `0xf1737317b461a21a72784232e7987adf95728e018566f030b3eb36b4623a59ec`

Read-only RPC check returned chain 0x1226, successful receipt at the recorded block, and runtime code matching the registry hash. Historical Legacy Testnet is preserved separately and is not a production target.

VERIFICATION STATUS

What has been tested—and what has not

62

Contract tests

Creation, shared funding, address roles, deadlines, review, payout, refunds, and reentrancy regressions.

4 + 5

RPC and browser checks

Four proxy guard tests and five local Playwright scenarios on disposable Hardhat. The latest run passed all five scenarios.

Build and types

Frontend build, frontend typecheck, deployment typecheck, and lint all passed.

Live wallet status

BridgeKey connection/signing and live bounty actions have not been checked. No bounty has been created or funded.

> Local E2E transactions are simulated engineering evidence. They are not Mainnet activity.

NEXT STEPS AND IMPACT

Ready for review; live actions stay gated

PatchPool gives maintainers and sponsors an inspectable way to pre-commit task terms, escrow, review authority, and settlement rules for software fixes.

For evaluation

Review the source, verify the deployment, and watch the clearly labeled local lifecycle demo.

Before public submission

Publish the reviewed source, confirm a hosted app, record a 2–3 minute video, and test every link from a signed-out browser.

Live BridgeKey signing, any Mainnet bounty transaction, public hosting, and hackathon submissions require separate manual action/approval. No finalist status is claimed.

Repository draft: github.com/ShravaniMankar664/PatchPool · Verified contract: MST Mainnet 4646